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Spatial Aware Peak Ranking in Mass Spectrometry Imaging

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***Abstract*—Mass spectrometry imaging (MSI) measures thousands of mass-to-charge (m/z) channels at every location of a tissue section, so analyses usually begin by selecting a small set of informative peaks. Learned peak-selection methods such as msiPL and S3PL rank m/z bins from unsupervised models, and S3PL adds spatial neighbourhood information through 3D convolution. This report asks whether cheap local spatial context can give a lightweight variational autoencoder (VAE) the same benefit, and which part of a learned pipeline actually determines peak quality. We fitted VAE variants with and without neighbourhood input, segmented their latent spaces with a Gaussian mixture model (GMM), and ranked peaks either by Integrated Gradients (IG) through the model or by direct correlation of each ion image with the GMM posterior maps. All methods were compared at matched peak budgets on 24 sections from three public collections (glioblastoma, colorectal adenocarcinoma and renal cell carcinoma) under predeclared decision rules, against reproduced msiPL and S3PL baselines and model-free rankings. Holding the segmentation fixed, posterior correlation outperformed IG in 136 of 140 paired comparisons across three training seeds, and this was statistically supported in all six dataset–model comparisons (Holm-adjusted $p \leq 0.047$). Neighbourhood input to the encoder gave no consistent, spatially specific gain, whereas spatially aware ranking statistics, namely segmentation correlation and Moran's I, were the strongest label-free rankings. A linear PCA segmentation performed comparably to the VAE. Finally, S3PL's published glioblastoma score lay outside the range of ten reproduced seeds; seeds altered S3PL mainly through weight initialisation, which switched individual sections between distinct solutions. The contribution is a controlled, statistically reported evaluation that bounds where spatial information helps learned MSI peak ranking.**

***Index Terms*—mass spectrometry imaging, peak selection, variational autoencoder, spatial context, Integrated Gradients, Moran's I, reproducibility**

I. INTRODUCTION

Mass spectrometry imaging (MSI) records a full mass spectrum at each location of a tissue section, mapping many molecules at once without choosing targets in advance [Caprioli et al. 1997; Schwamborn and Caprioli 2010]. The data are large: the sections used here have 972–4,761 measured pixels and 1,481–85,062 mass-to-charge (m/z) bins. Most analyses therefore begin by selecting a small set of informative peaks. *Peak picking* applies rule-based thresholds to each spectrum and can discard weak but informative ions [Gibb and Strimmer 2012; Lieb et al. 2020]. *Peak learning* instead fits an unsupervised model and ranks m/z bins by what the model uses. Examples are msiPL, a variational autoencoder (VAE) on single spectra [Abdelmoula et al. 2021], and S3PL, which adds spatial context through 3D convolution over tissue patches [Weigand et al. 2026a].

The proposal for this project, *Representation Learning for Autonomous Peak Detection in MSI*, set out to build *Spatial-msiPL*: an msiPL-style VAE given cheap local spatial context (the mean of a 3×3 neighbourhood), as a lightweight alternative to 3D convolution. Testing it fairly raised a second question: once a model is fitted, which part of the pipeline determines peak quality? The candidates are the representation, the segmentation built on it, and the attribution method used to rank bins.

We built Spatial-msiPL and its variants (uniform and attention-weighted neighbourhoods, larger windows, masked centres) and segmented each latent space with a Gaussian mixture model (GMM). Peaks were ranked by Integrated Gradients (IG) [Sundararajan et al. 2017] of the GMM posteriors, or by directly correlating each ion image with the posterior maps. All methods were scored at matched peak budgets under rules fixed before the results, on 24 sections from three public collections. The comparison included reproduced msiPL and S3PL baselines and model-free rankings.

Holding the segmentation fixed, correlation outperformed IG in 136 of 140 paired comparisons across three training seeds, and this was statistically supported in all six dataset–model comparisons. Neighbourhood input to the encoder gave no consistent, spatially specific gain. Spatially aware *ranking* statistics, namely segmentation correlation and Moran's I, were the strongest label-free rankings. A linear PCA segmentation matched the VAE. S3PL's published glioblastoma score lay outside ten reproduced seeds, and weight initialisation switched individual S3PL sections between distinct solutions.

The contributions are: (i) a predeclared, matched-budget evaluation of learned MSI peak ranking on three datasets; (ii) a ranking ablation, with paired statistics, showing that segment-then-correlate outperforms gradient attribution in this pipeline; (iii) controlled tests that bound where spatial information helps;

(iv) a statistical account of S3PL's reproduction gap and seed sensitivity. Section II reviews related work, section III states the research questions, section IV gives the methods, section V the results, section VI the discussion, and section VII concludes.

II. Background and Related Work

MSI data. An MSI data cube has two spatial dimensions and one m/z dimension [Buchberger et al. 2018]. Spectra are usually normalised by total ion current (TIC) [Deininger et al. 2011]. Neighbouring bins of one molecular peak are strongly correlated, so a bin-level evaluation can reward several bins of the same peak. This motivates the peak-level scoring used here.

Learned peak selection. msiPL trains a fully connected VAE [Kingma and Welling 2014] on single spectra and reads peaks from the first-layer weights [Abdelmoula et al. 2021]. massNet provides the GBM data used here [Abdelmoula et al. 2022]. S3PL convolves over a spectral patch, applies attention to the centre spectrum, reconstructs the whole patch, and selects the bins with the most attention [Weigand et al. 2026a]. It also introduced the mask-based spatial-correlation F1 score (mSCF1), and reported 0.496, 0.601 and 0.470 on GBM, CAC and RCC, against 0.403, 0.431 and 0.159 for msiPL. Other neural MSI work includes deep models of 3D data [Inglese et al. 2017] and ion-image embeddings [Guo et al. 2024; Weigand et al. 2026b]. Because S3PL changes the architecture, objective, normalisation and selector together, it is unclear which of them produces its gain, and whether much less spatial context would suffice.

Representation versus use. Reconstruction quality need not predict usefulness. Mazarura et al. [2025] found that reconstruction error, cluster quality and downstream classification ranked autoencoders, PCA and NMF differently on MSI data. We test the same distinction for peak ranking.

Attribution and segment-then-correlate. IG attributes a model output to its inputs by integrating gradients from a baseline to the input [Sundararajan et al. 2017]. Applied to a GMM posterior on a VAE latent space, it ranks bins through the full nonlinear encoder. The older route segments the tissue and correlates each ion image with the segments [Alexandrov et al. 2010; Alexandrov and Kobarg 2011]. Our pipeline contains both on the same segmentation.

Spatial statistics. Moran's I measures spatial autocorrelation [Moran 1950]. Spatial-structure measures have been used to judge MSI ion images [Alexandrov and Bartels 2013; Cassese et al. 2016], and spatially aware segmentation smooths features with neighbourhood weights before clustering [Alexandrov and Kobarg 2011].

Gap. Learned peak selectors are usually evaluated with unmatched peak counts, single training runs and no model-free references. It has not been tested under a common budget whether their advantage comes from spatial input, the representation or the ranking step, nor how stable it is across seeds.

III. Research Questions

The proposal asked: (P1) how local spatial covariance can be integrated into a 1D VAE without the overhead of 3D convolution; (P2) whether a spatial-coherence penalty with physical noise augmentation isolates true signal relative to spatially blind baselines; and (P3) whether this framework improves spatially structured peak identification over msiPL. The penalty, augmentation and Bayesian tuning did not pass their development gates. Fair scoring then showed that P3 depended more on the ranking step than on spatial input. The report therefore answers four questions:

RQ1. **Baseline.** Does the label-free VAE–GMM pipeline select better peaks than reproduced msiPL at the same budget? (P3, made fair.)

RQ2. **Ranking and representation.** Does the gain come from the attribution method (IG against correlation with the segmentation) or from the nonlinear encoder (VAE against PCA)?

RQ3. **Spatial context.** Does neighbourhood information give a consistent, spatially specific gain in the encoder, the segmentation or the ranking? (P1, P3.)

RQ4. **Reproducibility.** Do msiPL and S3PL reproduce, and how much do seeds change the results?

The full proposal-to-outcome table is in section A.

TABLE I
Datasets.

	GBM	CAC	RCC
Sections	8	8	8
Independent units	4 PDX lines	1 biopsy	8 subjects
m/z bins	85,062	1,481	10,200
Pixels per section	2,071–4,524	3,243–4,761	972–3,564
Classes (GMM K)	2	3	3

IV. Methodology

A. Data

We used the three collections on which S3PL was published (table I), so that both baselines could be reproduced. GBM is the massNet patient-derived xenograft (PDX) glioblastoma dataset (PR001292) [Abdelmoula et al. 2022]. CAC is the MetaboLights colorectal adenocarcinoma dataset (MTBLS415)[CHECK: acquisition paper]. RCC comes from Dill et al. [2010] via Cardinal [Bemis et al. 2015]. The expert masks come from the S3PL release [Weigand et al. 2026a].

- **GBM:** sections from the same PDX line are not independent, so GBM is also summarised per line.
- **CAC:** all sections come from one biopsy, so its comparisons are within one specimen.
- **RCC:** cancer, normal and an unannotated region (45–78% of pixels) are kept as three classes, with a cancer/normal-only sensitivity analysis.

Four sections (GBM108_positive, GBM22_2, 160TopL, 40TopL) were used for development. RCC was held out until the method was frozen.

B. Pipeline

Each section is modelled on its own spectra without labels, and its selected peaks are scored against its own masks. The

design is unsupervised and transductive. Masks never enter fitting or ranking, but they set two configuration values, which are disclosed below.

Spatial-msiPL. Following msiPL, spectra are TIC-normalised and a fully connected VAE (one hidden layer of 512 units, 5 latent dimensions) is trained for 100 epochs with Adam (learning rate 10^{-3}, batch size 128), using msiPL's categorical cross-entropy reconstruction loss plus KL divergence [Abdelmoula et al. 2021].

- The *centre-only* arm encodes one spectrum.
- The *uniform* arm concatenates it with the mean of its up to eight measured 3×3 neighbours.
- Both reconstruct the centre only.
- The variants are learned attention over neighbours (about 0.5% more parameters) [Vaswani et al. 2017], 5×5 to 9×9 windows, zeroed and shuffled neighbours, and centre masking.

Segmentation. A full-covariance GMM is fitted to the standardised latent means. Its number of components K equals the number of annotated classes, a class-informed choice that BIC did not support (section V-E).

Rankings. Each ranking orders the m/z bins, and the top K_{bin} are scored.

- *IG* [Sundararajan et al. 2017] integrates the gradient of the assigned GMM posterior from the section-mean spectrum to 12 sampled pixels per component (64 trapezoidal steps), and ranks bins by mean absolute attribution, balanced across components.
- *Posterior correlation* ranks bins by the absolute Pearson correlation of their ion images with each posterior map, balanced the same way, using the same GMM.
- *Model-free references* are mean intensity, variance, Moran's I over Moore-neighbour edges [Moran 1950], and random.
- *Ceiling:* a supervised oracle that correlates with the masks.

Test V replaces the VAE with 5-component PCA, keeping the GMM and correlation ranking. Test S smooths the latent means with structure-adaptive weights before the GMM [Alexandrov and Kobarg 2011].

Baselines. msiPL was reproduced and tuned to its published peak-count procedure. S3PL was run from its public repository with the released settings: 10 epochs, Adam (10^{-2}), batch size 16, and patch size p and peaks per patch z of 3/256 (GBM), 9/256 (CAC) and 5/1024 (RCC). The released code normalises each patch by its spatial maximum, while the paper describes TIC normalisation, so both were tested.

C. Evaluation and design

mSCF1. For each class mask and threshold $t \in \{0.3, 0.4, 0.5, 0.6\}$, a reference set of mask-correlated bins is formed using the S3PL evaluator's rank cutoff. The F1 score between the selected bins and the union of the reference sets is averaged over the four thresholds [Weigand et al. 2026a]. mSCF1 measures agreement with the masks, not chemical identity.

Budgets. All compared methods select the same number of bins, K_{bin}: the tuned msiPL count for GBM and CAC, and the released S3PL target for RCC. A peak-level scale (P1) collapses neighbouring bins of one mean-spectrum peak, so duplicates are not credited. On CAC and RCC, P1 is nearly singleton and acts as a second budget check.

Predeclared rules. Every comparison's rule was written before its outputs, in an append-only protocol.

- A difference "helps" if its mean is at least $+0.02$ mSCF1 and it is positive in at least 6/8 sections.
- Two methods are "equivalent" if the mean difference is within ± 0.02.
- Otherwise the label is "mixed", or "incomplete" if sections are missing.

All arms use training seed 1. The two core arms were retrained with seeds 2 and 3 on all sections (Test R). Seed results are given as mean $\pm$ SD across seeds of the seed-level collection means.

D. Statistics

The labels are accompanied by paired, per-section statistics:

- the exact two-sided Wilcoxon signed-rank test [Wilcoxon 1945];
- the Hodges–Lehmann (HL) median difference with an exact 95% CI [Hodges and Lehmann 1963];
- the matched-pairs rank-biserial correlation r.

Seeds are averaged within section first. A Mann–Whitney test was not used, because the comparisons are paired.

The 38 comparisons form five families by question: ranking (6), against legacy msiPL (4), encoder variants (17), segmentation (9) and S3PL (2). Holm correction [Holm 1979] is applied within each family. With $n = 8$ the smallest possible p is 0.0078, so only families of at most six tests can reach Holm significance. Effect sizes therefore carry the main weight.

A result is "statistically supported" only if its Holm-adjusted $p < 0.05$ *and* its practical label agrees. GBM is also summarised per PDX line ($n = 4$, smallest $p = 0.125$). Mixed models with random section and seed effects are descriptive.

E. Computation

Training ran on the University's Slurm cluster (RTX 3060 and 3090 GPUs), and ranking ran on CPU nodes.

- **Caching.** The original loader re-read every batch from HDF5 on shared storage. Data preparation took 24.6 s per batch against about 0.02 s of computation, so a 100-epoch GBM run took 11.3 h. Reading each section into memory once cut it to 23 min (30×), with identical losses and weights. The likely cause is repeated small network reads, but this was not separately measured.
- **Reuse of saved runs.** Same-seed reruns were checked to be bit-identical before any saved run was reused.
- **Timing.** Exact ranking equality depends on the CPU model, so timing runs used the processor type that produced each saved ranking.

V. Results

Scores are mSCF1 at K_{bin}, seed 1, bin level, paired by section, unless stated otherwise. Labels are quoted exactly.

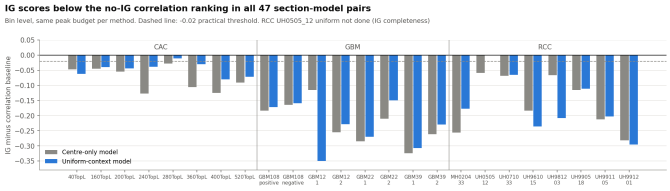


Fig. 1. IG minus posterior correlation per section and VAE arm (seed 1). All 47 valid comparisons are negative.

A. Baseline reproductions (RQ4)

Each baseline was run under its own protocol and evaluator.

- **msiPL** reproduced on all three collections: GBM 0.414 against 0.403 published, CAC 0.402 against 0.431, and RCC 0.157 against 0.159.
- **S3PL** reproduced on CAC (0.592 against 0.601) and on RCC (0.41–0.44 over three seeds, against 0.47).
- **S3PL on GBM was not reproduced:** 0.32 against 0.496, using the current public code and released settings. The paper's supplementary settings were not available (section V-F).

B. Correlation outperforms IG (RQ1, RQ2)

Table II scores every ranking on the same segmentation and budget.

Against msiPL. Both VAE–IG arms beat reproduced msiPL in every GBM and CAC section: 0.518 and 0.511 against 0.414 on GBM, and 0.528 and 0.560 against 0.402 on CAC. This is statistically supported in all four comparisons (HL $+0.09$ to $+0.16$, Holm $p = 0.031$). It compares whole pipelines, and one predeclared clause (budget stability) could not be assessed.

IG against correlation. With the GMM held fixed, correlation with the posterior maps beat IG in all 47 valid comparisons at seed 1 (figure 1). The predeclared rule's exact fall-through wording applies: "neither predeclared verdict applies, because IG scored below posterior-|PCC| in every section". Over seeds 1–3, correlation won 136 of 140 comparisons (table III).

- The strict per-seed rule failed only at CAC seed 2, where IG was below in 6/8 sections instead of the required 7/8, so the label is "does not hold overall".
- No collection mean ever reversed direction.
- With seeds averaged within section, the difference is statistically supported in all six dataset–arm comparisons (table III). Every section favours correlation ($r = -1$), and every raw p is the smallest attainable for its n.
- The mixed models agree (-0.24 [-0.27, -0.22] on GBM, -0.14 [-0.17, -0.12] on RCC) and attribute almost no variance to the seed.
- GBM is negative in every PDX line.
- Sampling 48 or 192 pixels for IG instead of 12 raised scores by at most 0.024.

Answer: the pipeline beats msiPL, but not because of IG. Given the same segmentation, direct correlation is better on all three datasets. The useful signal lies in the label-free segmentation.

C. Simple rankings and the metric

The best predeclared label-free ranking differs by dataset:

- GBM: posterior correlation (0.743);
- CAC: mean intensity (0.612), with correlation, Moran's I and variance all within 0.02;
- RCC: Moran's I (0.63).

CAC is near saturation: every method, including the oracle, converges near 0.41 at twice the budget. On GBM and RCC, sections whose GMM matches the masks better have correlation scores closer to the oracle (Spearman $\rho = -0.90$ and -0.84). mSCF1 rewards ion images that follow the masks, so mask-like segmentations score well when correlated against.

D. Spatial context (RQ3)

- **Uniform 3×3 context** "helps" IG on CAC ($+0.023$ over three seeds, 8/8 sections), but it is not statistically supported (Holm $p = 0.13$) and is 0.000 under correlation ranking. On GBM the result is "incomplete (7/8)", with a mean of $+0.000$.
- **It is not spatially specific.** On CAC, real and shuffled nonlocal neighbours tied (0.542 against 0.543). On GBM, zeroed context scored highest.
- **Larger windows** (7×7, 9×9) were 0.035 below 3×3 on GBM ("mixed") and "equivalent" on CAC.
- **Learned attention** was "equivalent" to uniform averaging ($+0.011$ GBM, -0.006 CAC). Its weights stayed near uniform with real neighbours (normalised entropy 0.98–0.999) but became selective with shuffled ones. The mechanism works; real neighbours are simply near-copies of the centre (median cosine about 0.98). Centre masking did not make attention selective and lowered GBM scores by 0.039.
- **Smoothed segmentation (Test S)** was never labelled "helps": four "equivalent", one "does not meet the 'helps' rule (labelled mixed)", and one "incomplete".

Answer: spatial input to the encoder or the segmentation gave no consistent, spatially specific gain. Spatial information helped as a ranking statistic, through correlation with coherent segments and Moran's I.

E. Is the nonlinear encoder needed? (RQ2)

Replacing the VAE with PCA-5 while keeping the GMM and correlation gave VAE $-$ PCA differences of $+0.023$ on GBM (5/8, "mixed"), -0.013 on CAC ("mixed") and "equivalent" on RCC. These were unchanged with the VAE averaged over three seeds. The VAE segmented slightly better (balanced accuracy 0.868 against 0.839 on GBM) without reliably better peaks.

PCA-5 also had *lower* reconstruction error than the VAE on all eight GBM sections, and higher error on 7/8 CAC sections, so reconstruction did not track peak quality [Mazarura et al. 2025]. BIC never validated the number of components: it chose the largest candidate ($K = 6$) every time.

F. S3PL: neighbours, reproduction and seeds (RQ3, RQ4)

Neighbour dependence. On CAC, retraining S3PL with every patch filled with the centre spectrum collapsed it from 0.591 to 0.18 and 0.15 (two seeds, 8/8 sections). Shuffled nonlocal neighbours recovered most of the score (0.55 against

TABLE II
COLLECTION MEAN MSCF1 AT K_{BIN} (SEED 1, BIN LEVEL). ALL RANKINGS EXCEPT THE BASELINES USE THE SAME GMM. BOLD: BEST LABEL-FREE RANKING. THE ORACLE USES THE MASKS AND IS A CEILING.

Coll.	VAE arm	IG	Post. corr.	Moran's I	Mean int.	Variance	msiPL	Random	Oracle
GBM	centre	0.518	**0.743**	0.684	0.483	0.497	0.414	0.009	0.809
GBM	uniform	0.511	**0.744**	0.684	0.483	0.497	0.414	0.009	0.809
CAC	centre	0.528	0.606	0.605	**0.612**	0.593	0.402	0.108	0.654
CAC	uniform	0.560	0.607	0.605	**0.612**	0.593	0.402	0.108	0.654
RCC	centre	0.412	0.568	**0.631**	0.498	0.508	—	0.171	0.756
RCC	uniform[a]	0.410	0.596	**0.635**	0.499	0.504	—	0.173	0.755

[a] 7 of 8 runs (UH0505_12 failed the predeclared IG completeness check). msiPL on RCC is not budget-matched.

TABLE III
IG — POSTERIOR CORRELATION ACROSS SEEDS 1–3 (BIN LEVEL): MEAN ± SD OF THE SEED-LEVEL COLLECTION MEANS, WINS FOR CORRELATION, AND PAIRED STATISTICS ON SEED-AVERAGED SECTIONS (HL WITH EXACT 95% CI; HOLM WITHIN F1).

	Mean ± SD	Corr. wins	HL [95% CI]	Holm p
GBM centre	-0.244 ± 0.018	24/24	-0.248 [-0.293, -0.186]	0.047
GBM uniform	-0.245 ± 0.011	23/23	-0.243 [-0.314, -0.159]	0.047
CAC centre	-0.077 ± 0.003	23/24	-0.078 [-0.116, -0.040]	0.047
CAC uniform	-0.055 ± 0.007	22/24	-0.046 [-0.080, -0.027]	0.047
RCC centre	-0.144 ± 0.011	22/23	-0.147 [-0.207, -0.082]	0.047
RCC uniform	-0.141 ± 0.041	22/22	-0.131 [-0.204, -0.056]	0.047

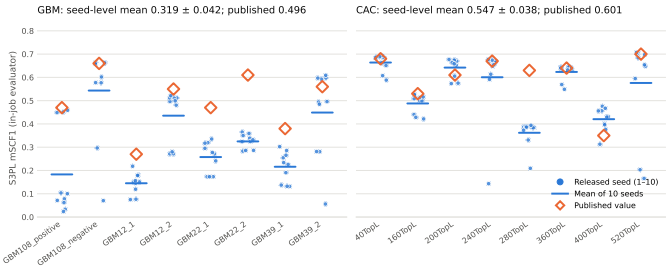


Fig. 2. S3PL mSCF1 per section over ten released seeds (dots), the seed mean (bar) and the published value (diamond), using S3PL's own evaluator.

0.59), so correct placement added only about $+0.04$, inconsistently ("mixed", Holm $p = 0.63$). Training for longer made S3PL worse: epoch 10 was best in 15/16 sections.

Reproduction gap. Over ten released seeds (figure 2), S3PL scored 0.319 ± 0.042 on GBM and 0.547 ± 0.038 on CAC.

- **GBM:** the published 0.496 is about 4 SD above our mean. It is above every one of our seeds in 6/8 sections, and best-of-ten per section reaches only 0.434. Labels: "outside our seed distribution" and "not explained by seed selection alone".
- **CAC:** the published 0.601 is just above our maximum (0.591), within its two-decimal rounding of best-of-ten (0.5998).
- **Normalisation:** switching to the TIC normalisation described in the paper did not raise GBM (0.303 against 0.324; HL -0.012 [-0.154, 0.099]), so it "does not explain" the gap.

Why seeds matter. About 52% of the score variance is between sections, 0–1% is a seed main effect, and 47% is a section $\times$ seed interaction.

- Seeds switch particular sections between distinct outcomes: GBM108_positive moves between 0.02 and 0.46, and CAC 520TopL between 0.17 and 0.71.
- The peak lists of different seeds can overlap by as little as 1%.
- Varying only the weight initialisation, or only the data order, showed that initialisation dominated (SD 0.222 against 0.001 on GBM108_positive), although order also mattered on GBM108_negative.
- Same-seed reruns were bit-identical.
- Under TIC the per-section seed SD fell from 0.061 to 0.011.

Answer: msiPL reproduces. S3PL reproduces except on GBM, where neither seeds nor normalisation explain the gap. Its starting weights select between distinct solutions, so single-seed comparisons against it are unreliable.

G. RCC and runtime

RCC. On the held-out RCC data, correlation beat IG in every valid run (0/8 and 0/7 positive), and Moran's I was the best label-free ranking. When only cancer and normal pixels are scored, the advantage shrinks: correlation beats IG in 5/8 runs (centre-only) and 6/7 (uniform). The three-class result therefore partly reflects separating tissue from the unannotated region.

Runtime. Given a segmentation, correlation took 1.1 s per GBM section against 770 s for IG (CAC 0.03 against 6.0 s; RCC 0.11 against 372 s), and every timed ranking reproduced the saved ranking exactly.

- Training: cached VAE runs took a median of 32–35 min (GBM), 1.3 min (CAC) and 2.2 min (RCC), against 13.7, 0.9 and 1.6 min for 10-epoch S3PL.
- Complete workflow, as the proposal planned to measure: on 160TopL the VAE took 122–135 s against 151–174 s for S3PL, with 1.2 against 1.3 GiB of RAM and 237 against 315 MiB of GPU memory.

These are native-protocol measurements, not an equal-work comparison.

VI. DISCUSSION

Why correlation outperforms IG. Three explanations fit.

- IG asks which bins would change a few sampled pixels' component probabilities. Correlation asks which ion images follow a component across the whole section, which is closer to what mSCF1 rewards.
- Sampling explains little: four- to sixteen-fold more IG pixels gained at most 0.024 against a gap of 0.05–0.25.
- mSCF1 favours mask-like segmentations.

Correlation is therefore an ablation finding about this pipeline: the label-free segmentation carries the signal, and gradient attribution through a dense VAE adds nothing to it. We do not claim that IG is generally worse, or that segment-then-correlate, which is established practice [Alexandrov and Kobarg 2011], is new.

What remains of the spatial hypothesis. In a dense VAE, no neighbourhood intervention beat shuffled, nonlocal context, and smoothing the segmentation did not help either. At 3×3 scale, neighbours are near-copies of the centre, so there is little for the model to use. Spatial information helped as a ranking statistic and in S3PL, which needs neighbour input. Even in S3PL, correct placement added only an inconsistent $+0.04$. Spatially aware *ranking*, not spatially aware *encoding*, is what the evidence supports, hence the title.

Reconstruction is not ranking quality. PCA-5 reconstructed GBM better than the VAE yet ranked peaks about equally well. Earlier pilots showed the same pattern: context models and the spatial penalty changed the latents without improving reconstruction. Reconstruction rewards fitting all intensity variance, while peak quality here depends on how well a low-dimensional segmentation separates the classes, in line with Mazarura et al. [2025].

The proposal's aim. The aim was autonomous, bias-free peak detection with a lightweight model. It is met only in part.

- Fitting and ranking are label-free, but the number of segments and the peak budget come from annotations, and BIC could not replace the first.
- The 1D model was cheaper than S3PL in time and memory on the section measured, but it did not benefit from its spatial input.

S3PL's instability means single-run comparisons against it can mislead. We report its GBM gap as unresolved and select no favourable seeds or checkpoints.

Limitations.

- **Statistical units.** GBM has four PDX lines, CAC one biopsy and RCC eight subjects. The p-values describe these sections, not a population.
- **Class-informed settings.** K and the peak budgets are class-informed.
- **The metric.** mSCF1 measures mask agreement, not molecular identity.
- **RCC** depends partly on its unannotated region.
- **Post hoc tests.** Tests S, V, N and R and the S3PL investigation were chosen after earlier results. Each was predeclared before its own outputs, and all outcomes are reported.
- **Untested.** Graph-based and ion-image spatial models, cross-section transfer and other IG settings were not tested.

VII. Conclusion

Under matched budgets and predeclared rules on three public collections, the results were as follows.

- Ranking peaks by correlation with a label-free GMM segmentation outperformed gradient attribution through a VAE. This was statistically supported and stable across seeds.
- Neighbourhood input to the encoder gave no consistent, spatially specific gain. Spatial information helped as a ranking statistic.
- A linear segmentation matched the VAE, and reconstruction did not track peak quality.
- msiPL reproduced. S3PL's GBM score did not, and its seeds switch sections between distinct solutions.

The contribution is a controlled evaluation rather than a new peak picker. It bounds where spatial information helps learned MSI peak ranking, and it shows how much conclusions depend on fair budgets, model-free references and seed replication.

Future work should:

- validate a label-free number of segments;
- test a predeclared combination of segmentation correlation and Moran's I on new data;
- evaluate graph- and ion-image-based spatial models;
- test transfer of fitted models between sections.

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Appendix A
Proposal commitments and outcomes

Appendix B
Per-seed results

Table V gives the per-seed IG minus posterior-correlation differences behind table III, at bin level. Each entry is the mean difference, with the number of sections in which IG was lower over the number of valid sections.

Appendix C
Further controls

The following supporting analyses are reported in full in the project repository and summarised here:

- the IG attribution-pixel test (12, 48 and 192 pixels per component);
- the frozen-model neighbour swaps and the out-of-distribution audit of shuffled inputs;
- the attention entropy and boundary analyses;
- the BIC curves and the BIC-K sensitivity;
- the CAC S3PL-versus-IG threshold and peak-overlap diagnostics;
- the Test S per-collection table;
- the PDX-line statistics for GBM ($n = 4$; all IG $-$ correlation differences negative, with the smallest attainable $p = 0.125$);
- the full 38-comparison statistics table, including P1 rows;
- the non-predeclared max-aggregated correlation variant (CAC 0.626–0.629).

[CHECK: choose which of these to show as figures or tables here]

Appendix D
Evidence, code availability and use of AI tools

Evidence and protocol. Every number in this report is traced to a committed result file. Each test's decision rule,

TABLE IV
PROPOSAL COMMITMENTS AND OUTCOMES.

Proposal commitment	Outcome	Reason for any change
Neighbourhood input to a 1D VAE (P1)	Built: uniform 3×3 mean, learned attention, 5×5–9×9 windows, centre masking	Delivered; extended with shuffled and zeroed controls to test spatial specificity
Spatial-coherence penalty on neighbouring latent means (P2)	Tested at $\lambda \in \{0, 0.01, 0.1, 1\}$; $\lambda = 0$ retained	Smoother latents but no gain in reconstruction, probe accuracy or Moran's I
Poisson noise augmentation, after DeepION (P2)	Tested; switched off	Did not pass its development gate
Bayesian optimisation of λ in $[0.01, 1]$	Omitted; a fixed grid was used	No penalty value passed the gate, so there was nothing to tune
Peak extraction from the L2 norm of first-layer weights	Kept as a comparator; replaced by GMM-targeted IG	First-layer weights ignore the nonlinear encoder; IG was later compared with direct correlation (RQ2)
Baselines: msiPL, S3PL, MALDIquant	msiPL and S3PL reproduced and audited; MALDIquant omitted; model-free rankings, random and supervised oracle added	Model-free rankings are a stronger and fairer reference than a spectrum-wise peak picker
Data: raw profile-mode imzML for GBM (PR001292) and CAC (MTBLS415)	Audited HDF5 conversions where needed; RCC added as a third, held-out dataset	Practical loading and a held-out confirmation set
mSCF1, PCC distributions, GPU memory and runtime	mSCF1 delivered with matched budgets, peak-level scoring and predeclared rules; memory and runtime measured; PCC distributions used in diagnostics	Bin-level scoring rewards duplicate bins of one peak
(Not in proposal)	Tests of the encoder (PCA), training seeds and S3PL neighbour topology	Added after reviews exposed missing controls

TABLE V
IG — POSTERIOR CORRELATION BY TRAINING SEED (BIN LEVEL).

Coll.	Arm	Seed 1	Seed 2	Seed 3
GBM	centre	-0.225 (8/8)	-0.261 (8/8)	-0.247 (8/8)
GBM	uniform	-0.233 (8/8)	-0.254 (8/8)	-0.248 (7/7)
CAC	centre	-0.078 (8/8)	-0.080 (8/8)	-0.074 (7/8)
CAC	uniform	-0.047 (8/8)	-0.057 (6/8)	-0.060 (8/8)
RCC	centre	-0.156 (8/8)	-0.138 (6/7)	-0.137 (8/8)
RCC	uniform	-0.185 (7/7)	-0.134 (8/8)	-0.104 (7/7)

deviations and execution log were written in an append-only protocol before the corresponding outputs existed. Same-seed reruns were checked for bit-identical results before saved runs were reused.

Code availability. The code, protocol, result files and figure scripts are available at [CHECK: repository URL and final commit hash]. The reproduced baselines use the public msiPL and S3PL repositories (S3PL v1.0.0, commit 660e3d9).

Use of AI tools. [CHECK: Zayd to complete according to the Wits AI declaration form: e.g. AI assistants were used to plan and review experiments, write and test analysis code under human review, check numbers against result files, and assist with drafting and editing text; all scientific decisions, approvals and final wording were the author's.]